

Surface:	Standard Equation
paraboloid:	$z = x^2 + y^2$ 
saddle:	$z = x^2 - y^2$ 
ellipsoid—sphere if uniformly scaled:	$x^2 + y^2 + z^2 = 1$ 
1-sheeted hyperboloid:	$x^2 + y^2 - z^2 = 1$ 
2-sheeted hyperboloid	$-x^2 - y^2 + z^2 = 1$ 
double-cone:	$z^2 = x^2 + y^2$ 

Example 1. Classify the surface by transforming it to a standard equation.

(a) $4x^2 - 16y^2 + z^2 = -16$

Solution. By noting the number of squared variables, we slim the choices above.

First we turn the constant term into 1 by dividing by:

Next we absorb positive coefficients into the square of the variables.

We perform substitutions $x/2 \mapsto x$ and $z/4 \mapsto z$ to execute a:

We perform swap $y \leftrightarrow z$ to execute a:

Hence, our surface is a:

(b) $(x + 3)^2 - y + z^2 = 1$

Solution. By noting the number of squared variables, we slim the choices above.

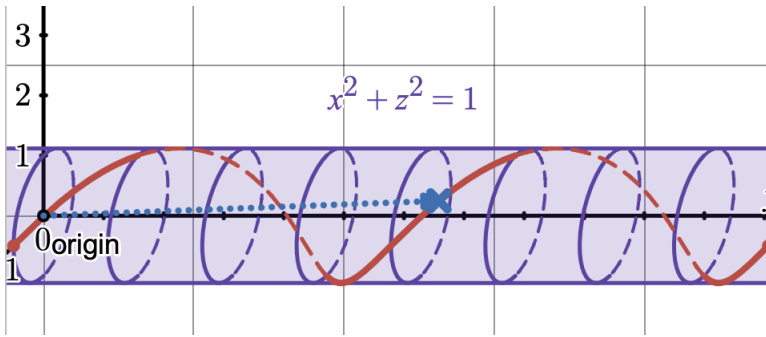
First we move the non-squared variable to the other side.

We perform substitutions $x + 3 \mapsto x$ and $y + 1 \mapsto y$ to execute a:

We perform swap $y \leftrightarrow z$ to execute a:

Hence, our surface is a:

A. **Curves.** The 1D objects in space are called **curves**.



This particular spiral is parametrized by the function:

$$\vec{r}(t) =$$

A single curve may have multiple parametrizations.

A twice as fast parametrization of this spiral is given by:

$$\vec{r}_{\times 2}(t) =$$

So, on the one hand, we have the fixed-in-space:

On the other hand, we have its different parametrizations, aka:

Example 2. Consider the curves $\mathcal{C}_1$ and $\mathcal{C}_2$ with respective parametrizations:

$$\vec{r}_1(t) = \langle t, 1 - t, 3 + t^2 \rangle$$

$$\vec{r}_2(t) = \langle 3 - t, t - 2, t^2 \rangle$$

Do the curves $\mathcal{C}_1$ and $\mathcal{C}_2$ intersect? If so, at what point?

Solution. This is a question about where the fixed-in-space curves intersect.

The curves have a point in common if there are values t_1, t_2 of parameters so:

We set this up as a system of equations:

Therefore, the point of intersection has position vector:

Example 3. Parametrize the curve of intersection of the cylinder $x^2 + y^2 = 1$ and the surface $z = xy$.

Solution. For clarity we view the curve in Desmos3D.

We select a parameter.

Points on the unit circle $x^2 + y^2 = 1$ may be located by their **polar** angle:

$$x =$$

$$y =$$

where:

The corresponding point on the surface $z = xy$ is:

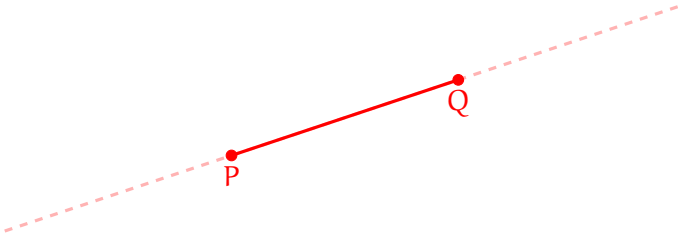
$$z =$$

All-in-all, if we use the polar angle as our parameter, a parametrization is:

$$\vec{r}(\theta) =$$

where:

Example 4. Parametrize the straight line segment from $P(2, 1, 3)$ to $Q(3, 4, 2)$.



Solution. We already know how to parametrize the line through P and Q .

$$\vec{r}(t) =$$

However, the segment does not consist of the entire line.

It only consists of the segment of the line starting at P and stopping at Q .

Observe that:

$$P =$$

$$Q =$$

Thus we restrict the domain of our parameter to:

B. **Path Derivatives.** We can differentiate a path, i.e. a parametrization.

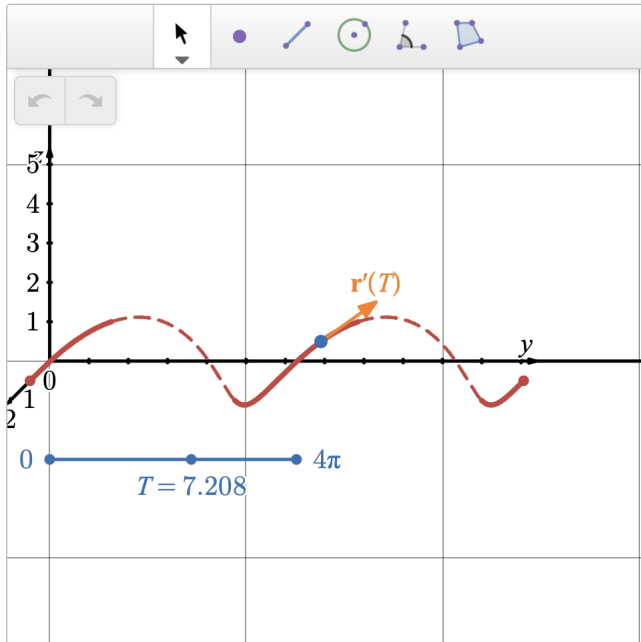
For example if:

$$\vec{r}(t) = \langle \cos t, t, \sin t \rangle$$

then its **path derivative** is:

$$\vec{r}'(t) =$$

When $\vec{r}(t)$ is interpreted as the path of a particle, we also call $\vec{r}'(t)$ the:



When rooted at the corresponding point on the curve, $\vec{r}'(t)$ points:

The length $\|\vec{r}'(t)\|$ of the velocity is the:

For example, the speed of the parametrization $\vec{r}(t) = \langle \cos t, t, \sin t \rangle$ is: